Solving Literal Equations — Set B

Easy level · 30 problems · A worked example comes before each type. Show every step.

Name: _____ Date: _____

Key Idea

A **literal equation** has more than one letter. To "solve for" one letter, get it alone on one side.

- Treat every *other* letter exactly like a number.
- Use the same moves as always: undo $+/-$ first, then undo $\times/\div$.
- **Ask yourself:** "If these letters were numbers, what would I do?" Then do it with the letters.

Common Mistake

Undo *everything* attached to your letter. From $d = rt$, dividing by r gives $t = \dfrac{d}{r}$, **not** $t = d - r$ (rt means multiply, so undo by dividing).

In a fraction answer, divide the *whole* side: from $ax = c - b$, $x = \dfrac{c-b}{a}$, **not** $c - \dfrac{b}{a}$.

Type 1 One Step: Add or Subtract

Watch Me — Worked Example

Solve $x - a = b$ for x.

If it were $x - 3 = 10$, you would add 3. Do the same with the letter.

Step 1 Add a to both sides $\quad x - a + a = b + a$

Answer $\quad \boldsymbol{x = b + a}$

1. Solve $x + 7 = y$ for x.

$x =$ _____

2. Solve $p + q = r$ for q.

$q =$ _____

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3. Solve $w - 6 = z$ for w.

$w =$ _____

4. Solve $a - c = b$ for a.

$a =$ _____

5. Solve $s + t = 20$ for s.

$s =$ _____

6. Solve $n - m = k$ for n.

$n =$ _____

Type 2 One Step: Multiply or Divide

Watch Me — Worked Example

Solve $A = lw$ for l.

lw means $l \cdot w$. Undo the multiplication by w.

Step 1 Divide both sides by w $\quad \dfrac{A}{w} = \dfrac{lw}{w}$

Answer $\quad \boldsymbol{l = \dfrac{A}{w}}$

7. Solve $A = bh$ for b.

$b =$ _____

8. Solve $d = rt$ for r.

$r =$ _____

9. Solve $P = 4s$ for s.

$s =$ _____

10. Solve $W = Fd$ for F.

$F =$ _____

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11. Solve $\frac{x}{5} = y$ for x.

$x =$ _____

12. Solve $\frac{m}{n} = p$ for m.

$m =$ _____

Type 3 Two Steps

Watch Me — Worked Example

Solve $2x + 3y = 6$ **for** y**.**

Get the y-term alone first, then undo its coefficient.

Step 1 Subtract $2x$ from both sides $\quad 3y = 6 - 2x$

Step 2 Divide both sides by 3 $\quad y = \frac{6 - 2x}{3}$

Answer $\quad \boldsymbol{y = \frac{6 - 2x}{3}}$

13. Solve $y = mx + b$ for b.

$b =$ _____

14. Solve $P = 2l + 2w$ for w.

$w =$ _____

15. Solve $3x + 2y = 12$ for y.

$y =$ _____

16. Solve $y = 2x + 7$ for x.

$x =$ _____

17. Solve $ax - b = c$ for x.

$x =$ _____

18. Solve $4x - y = 9$ for y.

$y =$ _____

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Type 4 Parentheses

Watch Me — Worked Example

Solve $a(x-b)=c$ for x.

Divide by the factor outside the parentheses, then undo what is inside.

Step 1 Divide both sides by a $\quad x-b=\dfrac{c}{a}$

Step 2 Add b to both sides $\quad x=\dfrac{c}{a}+b$

Answer $\quad \boldsymbol{x=\dfrac{c}{a}+b}$

19. Solve $2(x+a)=b$ for x.

$x=$ _____

20. Solve $5(m-n)=p$ for m.

$m=$ _____

21. Solve $c(x-d)=e$ for x.

$x=$ _____

22. Solve $3(a+b)=k$ for b.

$b=$ _____

23. Solve $P=2(l+w)$ for l.

$l=$ _____

24. Solve $a(y+2)=b$ for y.

$y=$ _____

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Type 5 Fractions and Formulas

Watch Me — Worked Example

Solve $V = \frac{1}{3}Bh$ for B.

Clear the fraction first by multiplying by its denominator.

Step 1 Multiply both sides by 3 $\quad 3V = Bh$

Step 2 Divide both sides by h $\quad \dfrac{3V}{h} = B$

Answer $\quad \boldsymbol{B = \dfrac{3V}{h}}$

25. Solve $A = \frac{1}{2}bh$ for h.

$h =$ _____

26. Solve $\dfrac{a+b}{2} = M$ for b.

$b =$ _____

27. Solve $V = \frac{1}{3}Bh$ for h.

$h =$ _____

28. Solve $y = \dfrac{x+4}{3}$ for x.

$x =$ _____

29. Solve $S = 2\pi rh$ for h.

$h =$ _____

30. Solve $F = \frac{9}{5}C + 32$ for C.

$C =$ _____

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